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EE Project-LBP

GJU-SATS

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Contents

1. Executive Summary	3
2. Company Context.....	4
3. Methodology: Analysis Techniques Used.....	5
4.Data & Assumptions.....	5
5. Economic Study	6
6. Risk Analysis: Sensitivity/Scenario Outcomes	6
Impact of Increased Maintenance Costs on Project Viability	6
Analysis Overview	6
Explanation of Inflation and MARR Effects.....	8
Breackeven Analysis.....	9
Tax Impact Calculation for LBP Packages.....	10
7. Conclusion & Recommendation.....	11
8. References.....	13

Lubrication best practices

1. Executive Summary

This project evaluates the economic and operational benefits of implementing **Lubrication Best Practices (LBP)** at **Hikma-Arab Medical Containers Co. (AMC)**, with the objective of reducing hydraulic system maintenance costs, decreasing lubricant consumption, and enhancing system reliability.

AMC currently spends approximately **96,250 JOD annually** on lubricant and component repairs due to contamination, inefficient oil change practices, and lack of proper handling systems. Oil sample analysis and inspection of damaged components confirmed high contamination levels, resulting in accelerated wear and frequent breakdowns.

Two alternative solutions were developed:

1. Basic LBP Package

Includes **basic technical training**, **installation of Des-Case breathers**, and **oil stations equipped with offline filtration units**. This package aims to reduce oil contamination, prevent cross-contamination, and improve oil handling practices.

- **Investment:** 49,615 JOD
- **Annual Savings:** 50,959 JOD/yearly
- **Payback Period:** ~1.1 years

Note: Good payback periods are generally short — often less than 2 years is preferred in many industries.

- **IRR:** ~99%
- **Note:** 99% isn't a huge number in predictive maintenance projects. As mentioned on the Emerson website: *Even though the costs of improving maintenance and reliability practices are not small, the benefits far outweigh the costs.* **ROI (Return on Investment) from 8:1 to 16:1** means that for every 1 unit of money spent, you get 8 to 16 units back in value or savings — a huge return. **IRR (Internal Rate of Return) from 50% to 250%** means that the annualized return rate of the investment is extremely high — much higher than most other investments — indicating that the investment pays back very quickly and continues to generate large returns over time.

- **Benefit–Cost Ratio:** 3.70

2. Advanced LBP Package

Includes **advanced technical training**, **RMF breathers with extended filtration life**, and **high-spec oil stations**. This package offers greater contamination control, higher downtime reduction, and longer component life extension.

- **Investment:** 86,863 JOD
- **Annual Savings:** 79,089 JOD/yearly
- **Payback Period:** ~1.26 years
- **IRR:** ~87%
- **Benefit–Cost Ratio:** 3.28

Economic analysis shows that while both proposals are highly profitable, the **Advanced LBP Package** offers higher total annual savings and greater long-term operational reliability. However, the **Basic LBP Package** has a slightly faster payback and higher IRR.

Recommendation: Proceed with the **Advanced LBP Package** if maximizing system reliability and long-term cost savings is the priority. The **Basic LBP Package** remains an attractive option if lower initial investment and faster ROI are preferred.

2. Company Context

Arab Medical Containers Co. (AMC), fully owned by Hikma Investment Company, is the leading manufacturer of healthcare-related plastics in Jordan. The company operates 50 production lines 24/7, producing over 150 different container types. AMC's manufacturing processes rely heavily on hydraulic systems for blowing and injection molding.

Current Problem

AMC experiences frequent failures of hydraulic proportional valves and pumps, with repair needs arising every 2–3 months. Oil contamination has been identified as the primary cause, with ISO 4406 cleanliness levels consistently falling below **OEM recommendations**. The current oil change practice is hour-based rather than condition-based, leading to both premature oil disposal and, at times, prolonged use of degraded oil.

Impact

- Annual lubricant costs: **≈12,250 JOD**
- Annual repair costs (valves + pumps): **≈84,000 JOD**
- Total annual cost: **≈96,250 JOD**
- Environmental impact from excessive waste oil disposal.

Scope

This project assesses the feasibility of implementing **Lubrication Best Practices (LBP)** to reduce costs, minimize failures, and improve environmental performance.

Objectives

1. Evaluate two feasible LBP implementation proposals:
 - **Basic LBP Package** (Basic Training + Des-Case breathers + oil stations).
 - **Advanced LBP Package** (Advanced Training + RMF breathers + upgraded oil stations).
2. Perform an economic analysis using **PW, IRR, Payback Period, FW, EUAW, and B/C ratio**.
3. Conduct sensitivity analysis to evaluate risk under varying conditions.
4. Recommend the most cost-effective and sustainable solution.

3. Methodology: Analysis Techniques Used

The following engineering economics methods were applied:

1. **Present Worth (PW)** – Calculates the value of all project cash flows discounted to present terms using a Minimum Attractive Rate of Return (MARR).
2. **Annual Worth (AW)** – Converts PW into an equivalent annual value for direct comparison.
3. **Future Worth (FW)** – Calculates total project worth at the end of the study period.
4. **Internal Rate of Return (IRR)** – Determines the effective return rate for each alternative.
5. **Payback Period (PBP)** – Measures the time needed to recover the initial investment from annual savings.
6. **Benefit–Cost Ratio (B/C)** – Compares total benefits to total costs to determine project profitability.

Calculations were conducted over a **5-year analysis period** using Microsoft Excel, applying actual AMC cost data and projected savings from the LBP implementation.

4. Data & Assumptions

Financial Data

- **MARR:** 12% (based on AMC's cost of capital).
- **Study period:** 5 years.

- **Currency:** Jordanian Dinar (JOD).

Cost Breakdown

- **Baseline:** Repairs = 84,000 JOD/year, Lube = 12,250 JOD/year.
- **Basic LBP Package:** Investment = 49,615 JOD; Annual Savings = 50,959 JOD.
- **Advanced LBP Package:** Investment = 86,863 JOD; Annual Savings = 79,089 JOD.

Equipment Lifespans

- Breathers: 0.5–1 year replacement cycle depending on model.
- Oil stations & filtration units: 10+ years.
- Training: Skills retained beyond study period; refresher training possible after 3 years.

Assumptions

- Savings are constant annually over the analysis period.
- No major inflationary changes for equipment costs.
- All savings are realized from Year 1 onward.

5. Economic Study

Proposal	Investment (JOD)	Annual Savings (JOD)	PW at 12% (JOD)	IRR	Payback (yrs)	B/C Ratio
Basic LBP	49,615	50,959	134,081	~99%	1.10	3.70
Advanced LBP	86,863	79,089	198,235	~87%	1.26	3.28

6. Risk Analysis: Sensitivity/Scenario Outcomes

Impact of Increased Maintenance Costs on Project Viability

Analysis Overview

1. Proposal 1: Limited Resilience to Cost Escalation

- **20% Maintenance Increase:**
 - **Payback Period:** Maintained After Year 1.
 - **IRR:** Dropped from **99% to 80%**
 - **Benefit-Cost Ratio:** Declined from **3.7 to 3.044** (still viable)

- **Total PV:** Reduced from **134,080.7905 JOD** to **101,428.72 JOD**.
- **50% Maintenance Increase:**
 - **Payback Period:** increases to EOY 3 – 4.
 - **IRR:** Dropped highly from **99%** to **49%**.
 - **Benefit-Cost Ratio:** dropped to **2.057**.
 - **Total PV:** Collapse to **52,447.02862 JOD**.

Interpretation:

While Proposal 1 remains technically viable at 50% cost increases, its extended payback period and significant PV reduction make it the **second-best option**. It works, but not optimally under stress.

2. Proposal 2: Robust Performance Under Pressure

- **20% Maintenance Increase:**
 - **Payback Period:** Unchanged After first year.
 - **IRR:** Slightly reduced from **93%** to **90%**.
 - **Benefit-Cost Ratio:** reduced from 3.480194526 to 3.377358618.
 - **Total PV:** Decreased.
- **50% Maintenance Increase:**
 - **Payback Period:** **Still held** (~1.26 years)
 - **IRR:** Moderately dropped to **85%**
 - **Benefit-Cost Ratio:** Stayed >2.0 with **3.223146257**
 - **Total PV:** Declined to **193,109.1533 JOD**.

Interpretation:

Proposal 2's **higher initial investment in durable components** (e.g., advanced breathers, training) provides **long-term cost stability**, making it resilient even to severe maintenance spikes.

Comparative Summary

Scenario	Proposal 1	Proposal 2
20% Cost Increase	Payback held, but IRR ↓ to by 19%	Payback & IRR stable

Scenario	Proposal 1	Proposal 2
50% Cost Increase	Payback increase; IRR < MARR (12%)	Payback held; IRR >> MARR (12%)
Cost Sensitivity	Moderate	Low

Recommendations

1. **Select Proposal 2** for long-term reliability. Its **resilience to cost shocks** ensures predictable returns.
2. **Limit Proposal 1** to scenarios with **guaranteed maintenance cost controls** (e.g., fixed-price contracts).
3. **Monitor IRR Thresholds:**
 - For Proposal 1, any maintenance increase **>30%** risks breaching MARR.
 - Proposal 2 remains safe up to **50%+ increases**.

Explanation of Inflation and MARR Effects

Inflation Overview

This study evaluates investment viability under inflation and varying MARR (Minimum Acceptable Rate of Return) scenarios using:

1. Inflation Forecasting:
 - Linear trend projection for short-term (5-year) inflation, validated by research (e.g., polynomial regression models).
 - Arithmetic mean of predicted years inflation to derive a general inflation rate (f).
2. Nominal MARR Calculation:
 - **Fisher Equation:**

$$\text{Nominal MARR} = (1 + \text{Real MARR}) \times (1 + f) - 1$$

- Tested across moderate (12–16%), and high (20%) real MARRs.

3. Scenario Testing:

Without inflation: Baseline performance at different MARRs.

With inflation: Adjusted nominal cash flows and discount rates to reflect real-world purchasing power erosion.

Inflation Effect

- Moving from **real MARR** to **nominal MARR** (adding inflation) raises the discount rate and reduces PW, B/C, and EUAW.
- **Example:** At a **real MARR of 12%**, PW for Proposal 1 drops from **134,081 JOD** (real) to **117,258 JOD** (nominal at 19.97%).
- This happens because the higher nominal rate discounts later cash flows more heavily.

MARR Effect (without inflation)

- Changing MARR without inflation shows only the impact of higher required returns on project viability.
- **Example:** For Proposal 1, at **12% MARR** PW is **134,081 JOD**, but at **16% MARR** it falls to **117,241 JOD**.
- The drop is purely due to higher discounting, not inflation.

Payback Period Changes

- Higher discount rates can affect the **Discounted Payback Period (DPP)** by making later cash flows contribute less to recovering the initial investment.
- **Example of unchanged payback:**
 - Proposal 1, at 12% real MARR: DPP after Year 1 – Year 2.
 - Proposal 1, at nominal 15.84%: DPP stays after Year 1 – Year 2
- **Example of late payback:**
 - Proposal 2, at 9% real MARR: Payback Period has changed to EOY4 - 5 Proposal 2, at real 9%: after year 4 – 5.

Breakeven Analysis

Interpretation and Decision Criteria

The main decision criterion applied here is the **Present Worth (PW) breakeven annual savings** for each proposal, using the **MARR of 12% over a 5-year period**. This criterion directly compares the time-value-adjusted benefits of each alternative against its initial investment cost.

The breakeven annual savings is the point at which an alternative's net present value equals zero — in other words, the minimum annual savings needed for the investment to be economically justified at the given MARR. We also calculated the extra annual savings required for Proposal B to be economically equivalent to Proposal A (pairwise breakeven), following the method in Chapter 11 of *Engineering Economy*.

The results show for **20% increase in maintenance costs**:

- **Proposal A:** Breakeven annual savings = **13,763 JOD/year**. Actual savings = **41,901 JOD/year** → comfortably above breakeven.

- **Proposal B:** Breakeven annual savings = **24,097 JOD/year**. Actual savings = **81,383 JOD/year** → also comfortably above breakeven.
- **Extra savings required for B to match A:** **10,333 JOD/year**. Actual difference = **39,482 JOD/year**, far exceeding this threshold.

The results show for **main the proposals:**

- **Proposal A:** Breakeven annual savings = **13,764 JOD/year**. Actual savings = **50,959 JOD/year** → comfortably above breakeven.
- **Proposal B:** Breakeven annual savings = **24,097 JOD/year**. Actual savings = **83,861 JOD/year** → also comfortably above breakeven.
- **Extra savings required for B to match A:** **10,333 JOD/year**. Actual difference = **32,902 JOD/year**, far exceeding this threshold.

Both proposals are economically acceptable under the 12% MARR. However, Proposal B offers a significantly higher PW, making it the financially stronger choice despite its higher initial cost. The large margin between actual and breakeven savings means the result is robust to moderate changes in costs or benefits.

It's important to note that the calculations **account for a 20% increase in maintenance costs**. This adjustment reduces the net annual savings for both alternatives, but even with this less favorable assumption, both still exceed their breakeven levels by a wide margin. This indicates strong economic resilience for both options, with Proposal B maintaining its advantage in total value creation.

Tax Impact Calculation for LBP Packages

Corporate Income Tax Rate: 20%

National Contribution Tax: 1%

Total Effective Tax Rate: 21%

Figures in Jordanian Dinar (JD)

If the investment is capital expenditure, it will be depreciated over 5 years.

1. Immediate Full Deduction Scenario (If the cost is fully deductible in the same year)

Package Type	Investment (JD)	Total Tax Rate	Tax Saving (JD)
Advanced LBP Package	86,836	21%	18,235.56
Basic LBP Package	49,615	21%	10,419.15

2. Depreciation Scenario (If depreciated equally over 5 years)

Summary:

- If the investment is fully deductible in the same year, the tax saving is **18,235.56 JD** for the Advanced package and **10,419.15 JD** for the Basic package.
- If depreciated over 5 years, the annual tax saving is **3,647.11 JD** for the Advanced package and **2,083.83 JD** for the Basic package.

Package Type	Investment (JD)	Depreciation Period	Annual Depreciation (JD)	Total Tax Rate	Annual Tax Saving (JD)
Advanced LBP Package	86,836	5 years	17,367.20	21%	3,647.11
Basic LBP Package	49,615	5 years	9,923.00	21%	2,083.83

Cash Flow (JOD)	Discount Factor	Present Value (JOD)	Cumulative PV (JOD)	Total PV (JOD)	MARR	DPP	IRR
-49615	1	-49615	-49615	134080.7905	Marr 12%	1.053336	104%
53042.83	0.892857143	47359.66965	-2255.33035		Marr 12%		
53042.83	0.797193878	42285.41935	40030.089		Marr 12%		
53042.83	0.711780248	37754.83869	77784.92769		Marr 12%		
53042.83	0.635518078	33709.67737	111494.6051		Marr 12%		
53042.83	0.567426856	30097.92626	141592.5313		Marr 12%		

Figure 1 Tax savings impact for the first proposal

7. Conclusion & Recommendation

Both the **Basic LBP** and **Advanced LBP** packages are financially viable and meet AMC's strategic objectives of cost reduction, downtime minimization, and sustainability improvement.

- **Basic LBP:** Recommended if the goal is minimal upfront investment with a quick ROI.
- **Advanced LBP:** Recommended if AMC prioritizes maximum savings, reliability, and environmental performance over the 5-year horizon.

Final Recommendation: Implement the **Advanced LBP Package** to achieve the highest total savings ($\approx 79,000$ JOD/year), superior contamination control, and long-term operational stability, while accepting a slightly longer payback period compared to the Basic package.

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